Pre-analysis plan: #ManyDaughters

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This pre-analysis plan (PAP) outlines the empirical strategy to test the following hypotheses:
H1. *Having daughters leads to stronger support for gender equality (as measured by attitudes).*
H2. *Having daughters leads to more left-leaning political preferences.*
H3. *Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).*
H4. *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).*

1 Sample inclusion & exclusion criteria

Restrictions for the estimation sample are the following:

- Only individuals with valid identification data, so valid HID and PID/PERSNR;
- Only individuals who have individual level data, NETTO between 10-19;
- Only individuals who have at least one child for whom there is information, so for whom BIOVALID = 31 at least once, or, starting from 2015, also BIOVALID = 51;
- Only individuals 18 or older; check their age using GEBJAHR, GEBMONAT, PGPIYEAR, PGMONTH;
- Only individuals with valid numbers for the number of children SUMKIDS;
- Only individuals whose children's sex has been recorded, so valid KIDSEXnn for all children;

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- Waves between 2001–2020 inclusive, as in 2000 there was no information about one's children for men. For each hypothesis, I will additionally limit the sample to the years when the outcome variable has been collected;
- In case there are missing values for any of the necessary variables for a hypothesis to be tested, I will do a listwise deletion.

I assume that the gender composition of existing children doesn't influence the stopping rule (stopping to have children once a certain gender composition is achieved). Also, SUMKIDS and KIDSEXnn generally refer to biological children, so I will focus on them (excluding adopted, foster or stepchildren).

Additional restriction for H3: Only two-parent households with one man and one woman, both appearing in the dataset (with valid partner ID PGPARTNR and PGPARTZ values 1 or 2).

2 Outcome variables

- **H1**: PLJ0621–25 (views about women's position, measured 2016–2018). Constructed via PCA from these five variables.
- **H2**: PLH0333 (party vote choice, measured 2014 & 2018). Recoded to binary: 1 if AfD (value 37), 0 otherwise (excluding ineligible [29] and non-voters [28]).
- **H3**: Weekly leisure hours of man divided by total leisure hours of both partners, using PLI0051, PLI0036, PLI0010.
- **H4**: Same as H1; model estimated separately for men and women.

3 Independent variables

Included in all models (unless otherwise specified):

- Variable of interest: number of daughters (constructed from SUMKIDS and KIDSEXnn)
- Number of children (SUMKIDS)
- Age in months (from GEBJAHR, GEBMONAT, PGPIYEAR, PGMONTH)
- Years of education (PGBILZT)

- Household income (HGHINC; logged)
- East/West Germany indicator (BGSAMPREG)
- Employment status dummy (fulltime: PGEMPLST = 1)
- Disability status dummy (PLE0040 = 1)
- Marital status dummies (PGFAMSTD: single, widowed, divorced, separated; reference = married)
- Migration background dummies (MIGBACK: no migration background [reference], direct, indirect)
- Federal state dummies (HGNUTS1)
- Survey year fixed effects (SYEAR)
- Sex (SEX) will be used to show split sample regressions for men and women (for H2 and H4), even if this is not explicitly required for H2.

4 Estimation

I will use the model specified in Oswald and Powdthavee (2010) in Table 8, and add two dummies for the migration background:

$$\begin{aligned} Outcome_{it} = \alpha_i + \gamma_t + \beta_1 \cdot NumDaughters_{it} + \sum_{k=1}^{15} \beta_{2k} \cdot \mathbf{1}[NumChildren_{it} = k] \\ + \beta_3 \cdot \frac{Age_{it}^2}{100} + \beta_4 \cdot \log(HHIncome_{it}) + \beta_5 \cdot YrsSch_{it} \\ + \beta_6 \cdot EmployedFT_{it} + \beta_7 \cdot Disabled_{it} + \beta_8 \cdot Single_{it} \\ + \beta_9 \cdot Widowed_{it} + \beta_{10} \cdot Divorced_{it} + \beta_{11} \cdot Separated_{it} \\ + \beta_{12} \cdot EastGermany_{it} + \beta_{13} \cdot DirMigBack_i \\ + \beta_{14} \cdot IndirMigBack_i + \gamma_{rt} \end{aligned} \tag{1}$$

The model includes individual fixed effects (α_i), year fixed effects (γ_t), number of daughters (NumDaughters_{it}), categorical dummies for number of children (NumChildren_{it}, reference = no children), a quadratic term for age ($\text{Age}_{it}^2/100$), log household income ($\log(\text{HHIncome}_{it})$), years of education ($\text{YrsSch}_{it}$), and dummies for full-time employment (EmployedFT_{it}), disability (Disabled_{it}), marital status ($\text{Single}_{it}, \text{Widowed}_{it}, \text{Divorced}_{it}, \text{Separated}_{it}$; reference = married), migration background ($\text{DirMigBack}_i, \text{IndirMigBack}_i$; reference = no migration background) and

living in East Germany (EastGermany_{it}), as well as federal state fixed effects ($\gamma_r t$).

For H3 only: I will run one regression per household. I will drop marital status dummies and use the rest of individual level independent variables for each partner.

5 Testing & interpretation

For H1, H3 and H4 I will use OLS regressions with individual fixed effects. For H2, I will use a Logit regression with individual fixed effects and report log-odds coefficients. I will cluster standard errors at the individual level using option *vce(cluster id)*.

The coefficient of interest for H1, H2 and H4 is β_1. Evidence in favor of H1 is that β_1 is positive when the outcome variable increases with more liberal views on women's position. Evidence in favor of H2 is a negative β_1. Evidence in favor of H3 is a t-test showing that $|\alpha_i + \beta_i - 0.5| < |\alpha_i - 0.5|$. Evidence in favor of H4 is a t-test showing that β_1 is higher in the subsample of men than in the subsample of women. I will interpret p-values of $< 5\%$ as significant.

References

Oswald, A. J. and Powdthavee, N. (2010). Daughters and left-wing voting. *Review of Economics and Statistics*, 92(2):213–227.